Module 1

P3: Exploring data: Displaying datasets using PROC PRINT (SAS), Data Viewer (SPSS), and View() or print() (R).

In console type

data <- read.csv(file.choose())

Then choose the data.csv file

CODE:

```
View(data)

str(data)

head(data)

tail(data)

plot(data$Age, data$Salary,

    main = "Age vs Salary",

    xlab = "Age",

    ylab = "Salary",

    pch = 8,

    cex = 2,

    col = "blue")
```

P5: Sorting data using PROC SORT in SAS, Sort Cases in SPSS, and arrange() in R.

Code:

```
data<-data.frame(

  Name=c("Ayesha","Bob","Charlie","David","Eva"),

  Age=c(20,30,25,31,22),

  Salary=c(70000,65000,44000,40000,55000)

)

print("Original Data")

print(data)
```

```
sorted_age <- data[order(data$Age), ]
print("Sorted by Age (Ascending):")
print(sorted_age)

sorted_age <- data[order(-data$Age), ]
print("Sorted by Age (Descending):")
print(sorted_age)

data$Dept<-c("IT","HR","Finance","IT","HR")
sorted_multi<-data[order(data$Dept,data$Age),]
print("Sorted by Dept Than Age")
print(sorted_multi)
```

P6: Combining and appending datasets using MERGE in SAS, Merge Files in SPSS, and merge() or bind_rows() in R.

CODE:

```
data1 <-data.frame(
  ID =c(1,2,3),
  Name = c("Ayesha","Bob","Charlie")
)
data2<-data.frame(
  ID = c(2,3,4),
  Salary = c(50000,60000,45000)
)
merged_data <-merge(data1,data2,by = "ID",all = TRUE)
print("Merged Data (by ID):")
print(merged_data)
```

```
data3<-data.frame(
  ID=c(5,6),
  Name=c("David","Eva")
)
append_data<-rbind(data1,data3)
print("Append Data")
print(append_data)
```

P8: Applying basic data cleaning functions: handling missing values using MISSING (SAS), Define Missing Values (SPSS), and na.omit()/replace_na() in R.

CODE:

```
library(tidyr)

data <- data.frame(
  Name = c("Ayesha","Saamiya","Saba","Mantasha"),
  Age = c(25, NA, 7, NA),
  Salary = c(40000, 55000, NA, 48000)
)

print("Ayesha 17")
print("Original Data with Missing value:")
print(data)

clean_data <- na.omit(data)
print("After Removing Missing Values:")
print(clean_data)

filled_data <- replace_na(data, list(Age = 0, Salary = 0))
```

```
print("After Replacing Missing Values:")
print(filled_data)
```

P10: Creating new variables using transformations and calculations (all three software).

CODE:

```
data <- data.frame(
  Name = c("Ayesha","Bob","Saba"),
  Salary = c(40000, 55000,48000),
  Age = c(25, 32, 28)
)

print("Original Data: ")
print(data)

data$Bonus <- data$Salary*0.10

data$Category <-ifelse(data$Age>=30, "Senior", "Junior")

data$TotalIncome <-data$Salary + data$Bonus

print("Data with New Variables:")
print(data)
```

P13: Identifying and handling duplicates using PROC SORT NODUPKEY (SAS), Identify Duplicate Cases (SPSS), and distinct() (R).

```
url <-
"https://raw.githubusercontent.com/Statology/Miscellaneous/main/basketball_data.csv"

data <- read.csv(url)
```

```
cat("Original Dataset:\n")
print(head(data))

library(dplyr)
dup_data <- rbind(data, data[1, ])
duplicate_rows <- dup_data[duplicated(dup_data), ]
cat("\nDuplicate Rows:\n")
print(duplicate_rows)
unique_data <- distinct(dup_data)
cat("\nAfter Removing Duplicates:\n")
print(head(unique_data))
```

P14: Extracting date components using DATE functions (SAS), Date & Time Wizard (SPSS), and lubridate:: functions (R).

```
url <-
"https://raw.githubusercontent.com/Statology/Miscellaneous/main/basketball_data.csv"

data <- read.csv(url)
print("Ayesha 20")

cat("Original Dataset:\n")
print(head(data))

library(lubridate)

data$Date<-as.Date("2024-01-01")+0:(nrow(data)-1)

data$Year<-year(data$Date)
```

```
data$Month<-month(data$Date)

data$Day<-day(data$Date)

cat("\nDate Components Extracted (First 6 rows:\n")

print(head(data))
```

P15: Generating basic summaries using PROC CONTENTS (SAS), Variable View (SPSS), and str() or summary() (R).

```
url <-
"https://raw.githubusercontent.com/Statology/Miscellaneous/main/basketball_data.csv"


data <- read.csv(url)

cat("Original Dataset:\n")

print(head(data))

library(lubridate)

data$Date<-as.Date("2024-01-01")+0:(nrow(data)-1)

data$Year<-year(data$Date)

data$Month<-month(data$Date)

data$Day<-day(data$Date)

cat("\nDate Components Extracted (First 6 rows:\n")

print(head(data))
```

MODULE 2

P2: Generating frequency tables using PROC FREQ (SAS), Frequencies (SPSS), and table() or count() (R).

```
library(ggplot2)

data<-diamonds


print("Original Data(First 6 Rows):")

print(head(data))
```

```
print("Frequency Table for Cut")
freq_cut <-table(data$cut)
print(freq_cut)
library(dplyr)
print("Frequency Table using Count():")
count_cut <-data%>%count(cut)
print(count_cut)
```

P4: Performing one-sample t-tests using PROC TTEST (SAS), T-Test (SPSS), and t.test() (R).

```
data(mtcars)
print("Original Data(First 6 Rows):")
print(head(mtcars))
print("One-Sample t-Test on MPG (mu=20):")
one_sample_test<-t.test(mtcars$mpg, mu=20)
print(one_sample_test)
```

P7: Performing one-way ANOVA using PROC ANOVA (SAS), One-Way ANOVA (SPSS), and aov() (R).

```
data("iris")

print("Iris Dataset (First 6 rows):")
print(head(iris))

one_way_anova <- aov(Sepal.Length ~ Species, data = iris)

print("One-Way ANOVA Results:")
summary(one_way_anova)
```

```
print("Mean Sepal Length by Species:")
print(aggregate(Sepal.Length ~ Species, data = iris, mean))



anova_p_value <- summary(one_way_anova)[[1]][["Pr(>F)"]][1]


if (anova_p_value < 0.05) {
  print("Conclusion: There is a significant difference in sepal length among species.")
} else {
  print("Conclusion: There is no significant difference in sepal length among species.")
}
```

P9: Conducting Chi-square tests using PROC FREQ (SAS), Crosstabs with Chi Square (SPSS), and chisq.test() (R).

```
library(ggplot2)
data <- diamonds


print("Diamonds Dataset (First 6 Rows):")
print(head(data))


chi_table <- table(
  data$cut,
  data$color
)


print("Contingency Table (Cut vs Color):")
print(chi_table)


chisq_test <- chisq.test(chi_table)
```

```
print("Chi-Square Test Results:")
print(chisq_test)

if (chisq_test$p.value < 0.05) {
  print("Conclusion: There is a significant association between diamond cut and color.")
} else {
  print("Conclusion: There is no significant association between diamond cut and color.")
}
```

P10: Creating graphical reports using PROC REPORT (SAS), Chart Builder (SPSS), and ggplot2 (R).

```
library(ggplot2)
data(iris)
sepal_plot <- ggplot(
  iris,
  aes(x = Species, y = Sepal.Length)
) +
  geom_bar(
    stat = "summary",
    fun = mean,
    fill = "skyblue"
  ) +
  labs(
    title = "Mean Sepal Length By Species",
    x = "Species",
    y = "Sepal Length"
  )
print(sepal_plot)
```

P11: Generating histograms and box plots using PROC SGPLOT (SAS), Graphs (SPSS), and ggplot2 (R).

```
library(ggplot2)
data(iris)
print("Iris Dataset(First 6 Rows)")
print(head(iris))
hist_plot<-ggplot(
  iris,
  aes(x=Sepal.Length)
)+
  geom_histogram(
    fill="lightblue",
    color="black",
    bins=10
  )+
  labs(
    title="Histogram of Sepal Length",
    x="Sepal Length",
    y="Frequency"
  )+
  theme_minimal()
print(hist_plot)
box_plot<-ggplot(
  iris,
  aes( x=Species, y=Sepal.Length)
)+
  geom_boxplot(
    fill="lightgreen"
```

```
  )+
  labs(
    title = "Box Plot of Sepal Length by Species",
    x="Sepal Length",
    y="Frequency"
  )+
  theme_minimal()
print(box_plot)
```

P12: Generating correlation matrices using PROC CORR (SAS), Correlation (SPSS), and cor() (R).

```
data(iris)
print("Iris Dataset (First 6 rows)")
print(head(iris))
numeric_data<-iris[,1:4]
print("Selected Numeric Variables:")
print(head(numeric_data))
cor_matrix<-cor(
  numeric_data,
  method="pearson"
)
print("Correlation Matrix:")
print(round(cor_matrix,2))
```

p13: Performing linear regression analysis using PROC REG (SAS), Regression (SPSS), and lm() (R).

```
data("mtcars")
print("MTCARS Dataset (First 6 Rows):")
print(head(mtcars))
linear_model <- lm(mpg ~ wt, data = mtcars)
```

```
print("Linear Regression Summary:")
summary(linear_model)

plot(
  mtcars$wt,
  mtcars$mpg,
  main = "Linear Regression: MPG vs Weight",
  xlab = "Car Weight (1000 lbs)",
  ylab = "Miles Per Gallon (MPG)",
  pch = 16
)

abline(linear_model, lwd = 2)
```